

KEY



MATH 241-Spring 2014

Name : _____

QUIZ 1

Student ID : _____

04/03/2014

Section : _____

1. Let
$$\left[\begin{array}{cccc|c} k+1 & 1 & 2 & 0 & 2 \\ 0 & k & -1 & -3 & k+1 \\ 0 & 0 & k-1 & 0 & -1 \\ 0 & 0 & 0 & k-3 & k+2 \end{array} \right]$$

be the augmented matrix of the system $AX = B$

(a) Determine the values of k for which the system has a unique solution, infinitely many solutions, no solution.

(b) Find the solution of the system when $k = 0$.

a). If $k=1$, 3rd row of the matrix gives, $0=-1$. Hence there is no soln.

If $k=3$, 4th row gives $0=5$. Again there is no soln.

If $k=-1$,
$$\left[\begin{array}{cccc|c} 0 & 1 & 2 & 0 & 2 \\ 0 & -1 & -1 & -3 & 0 \\ 0 & 0 & -2 & 0 & -1 \\ 0 & 0 & 0 & -4 & 1 \end{array} \right] \xrightarrow{R_1+R_2} \left[\begin{array}{cccc|c} 0 & 1 & 2 & 0 & 2 \\ 0 & 0 & 1 & -3 & 0 \\ 0 & 0 & -2 & 0 & -1 \\ 0 & 0 & 0 & -4 & 1 \end{array} \right] \xrightarrow{2R_2+R_3} \left[\begin{array}{cccc|c} 0 & 1 & 2 & 0 & 2 \\ 0 & 0 & 1 & -3 & 0 \\ 0 & 0 & 0 & -6 & -1 \\ 0 & 0 & 0 & -4 & 1 \end{array} \right]$$

$$\left[\begin{array}{cccc|c} 0 & 1 & 2 & 0 & 2 \\ 0 & 0 & 1 & -3 & 0 \\ 0 & 0 & 0 & -6 & -1 \\ 0 & 0 & 0 & -4 & 1 \end{array} \right] \xrightarrow{-\frac{2}{3}R_3+R_4} \left[\begin{array}{cccc|c} 0 & 1 & 2 & 0 & 2 \\ 0 & 0 & 1 & -3 & 0 \\ 0 & 0 & 0 & -6 & -1 \\ 0 & 0 & 0 & 0 & \frac{5}{3} \end{array} \right]$$

last row gives $0=\frac{5}{3}$

Hence, there is also no soln. in the case $k=-1$.

If $k=0$,
$$\left[\begin{array}{cccc|c} 1 & 1 & 2 & 0 & 2 \\ 0 & 0 & -1 & -3 & 1 \\ 0 & 0 & -1 & 0 & -1 \\ 0 & 0 & 0 & -3 & 2 \end{array} \right] \xrightarrow{-R_2+R_3} \left[\begin{array}{cccc|c} 1 & 1 & 2 & 0 & 2 \\ 0 & 0 & -1 & -3 & 1 \\ 0 & 0 & 0 & 3 & -2 \\ 0 & 0 & 0 & -3 & 2 \end{array} \right] \xrightarrow{R_3+R_4} \left[\begin{array}{cccc|c} 1 & 1 & 2 & 0 & 2 \\ 0 & 0 & -1 & -3 & 1 \\ 0 & 0 & 0 & 3 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

$$\left[\begin{array}{cccc|c} 1 & 1 & 2 & 0 & 2 \\ 0 & 0 & -1 & -3 & 1 \\ 0 & 0 & 0 & 3 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

we have 4 unknowns, 3 nonzero eqns, hence there are infinitely many solns.

b) Continuing from the last part of a)

$$+2 \left[\begin{array}{cccc|c} 1 & 1 & 2 & 0 & 2 \\ 0 & 0 & -1 & -3 & 1 \\ 0 & 0 & 0 & 3 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

gives

$$\begin{aligned} x+y+2z &= 2 \\ -2-3w &= 1 \\ 3w &= -2 \end{aligned}$$

backsub

$$\Rightarrow \begin{cases} w = -2/3 \\ z = 1 \\ y = t \\ x = -t \end{cases}$$

Soln. in the case $k=0$

Note: A student who solves only part (b) in the correct manner, should get $2+2=4$ pts.